

EROLLA RISHVIN REDDY

IoT & Embedded Systems Engineer • IoT Security • Smart Systems

ESP32 • Arduino • Embedded C/C++ • MQTT • Python • IoT • Sensors • Edge Computing • IoT Security

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GitHub

Portfolio

LinkedIn

Professional Summary

Computer Science Engineering student specializing in **Blockchain, IoT & Cybersecurity (BIC)** with a focused interest in **IoT Engineering, Embedded Systems, IoT Security, Edge Computing, and Smart Connected Systems**. Experienced in designing and prototyping **hardware-software integrated IoT solutions** using **ESP32, Arduino, Embedded C/C++, Python, MQTT, sensors, actuators, wireless communication, REST APIs, and real-time telemetry**. Hands-on experience across the complete IoT lifecycle including **sensor integration, microcontroller programming, device communication, data acquisition, edge processing, cloud connectivity, automation, monitoring, and cybersecurity**. Co-inventor of a **Government of India registered IoT design patent** and developer of IoT systems spanning **smart irrigation, smart waste management, smart city infrastructure, biometric systems, and blockchain-based IoT forensics**. Strong foundation in **IoT architecture, embedded programming, computer networks, secure communication, database systems, system design, and problem solving**.

Education

Woxsen University

B.Tech in Computer Science and Engineering

Specialization: Blockchain, IoT & Cybersecurity (BIC)

Hyderabad, India

2024 – 2028

CGPA: **9.01/10**

Technical Skills

IoT & Embedded Systems: Internet of Things (IoT), Embedded Systems, ESP32, Arduino Uno, Embedded C/C++, Microcontroller Programming, Hardware-Software Integration, Sensor Integration, Actuator Control, IoT Prototyping, Real-Time Monitoring

Sensors & Hardware: Soil Moisture Sensors, Temperature Sensors, Phototransistors, Ultrasonic Sensors, Gas Sensors, Weight Sensors, RFID, Fingerprint Sensors, DC Motors, Water Pumps, Servos, LCD/I2C Displays

IoT Communication: MQTT, Wi-Fi, Bluetooth/BLE, HTTP/HTTPS, REST APIs, Device-to-Device Communication, Device-to-Cloud Communication, Telemetry, Real-Time Data Transmission

Embedded & Programming: Embedded C/C++, Python, TypeScript, JavaScript, Java, SQL, Microcontroller Programming, Serial Communication, Hardware Control, Data Acquisition

IoT Security: IoT Security, Embedded Security, Secure Device Communication, Authentication, Encryption, Cryptographic Hashing, Network Security, Threat Modeling, Digital Forensics, Evidence Integrity

Backend & Cloud: Node.js, FastAPI, PostgreSQL, MySQL, MongoDB, Firebase, AWS, RESTful APIs, Backend Development, Database Integration, Real-Time Data Processing

Edge AI & Intelligent Systems: Edge Computing, Machine Learning, Predictive Analytics, XGBoost, Computer Vision, AI Integration, Intelligent Automation, Route Optimization

IoT Development Tools: Arduino IDE, Tinkercad, Git, GitHub, GitHub Actions, Docker, Linux, Postman, Antigravity IDE

Engineering Fundamentals: IoT Architecture, System Design, Data Structures & Algorithms, Computer Networks, Operating Systems, Database Management Systems, Object-Oriented Programming, Software Engineering, Problem Solving

Patent & Innovation

Government of India Registered Design Patent – IoT Connectivity Device *Co-Inventor — Design No: 470097-001*

- Co-inventor of a **Government of India registered design patent** for an IoT connectivity device, demonstrating applied innovation in **Internet of Things, connected-device engineering, and smart infrastructure**.
- Contributed to the design of a **compact IoT connectivity solution** supporting device communication, connectivity, and integration within connected-device ecosystems.
- Recognized through **Design Registration No. 470097-001**, demonstrating experience translating an engineering concept into a formally registered **intellectual property (IP) asset**.

Technical Experience

Pega Platform Intern – Pegasystems

Aug 2026 – Present

- Selected for the **National Internship Program 2026 with SmartBridge**, gaining hands-on exposure to enterprise application development using the **Pega Platform**.
- Developing practical knowledge of **Business Process Management (BPM)**, **Case Management**, **workflow automation**, **Digital Process Automation (DPA)**, and **low-code application development**.
- Applying structured application-development practices covering **business process modeling**, **case lifecycle management**, **data integration**, **application configuration**, and **enterprise workflow design**.

Founder & IoT Systems Engineer – Rishvin Labs

2025 – Present

- Founded and lead **Rishvin Labs**, an independent engineering initiative developing systems across **IoT**, **embedded systems**, **cybersecurity**, **digital forensics**, **blockchain**, **automation**, and **smart connected technologies**.
- Designed and prototyped **IoT and embedded systems** integrating microcontrollers, sensors, actuators, wireless communication, real-time telemetry, backend services, and security controls.
- Worked with **ESP32**, **Arduino**, **Embedded C/C++**, **Python**, **MQTT**, **TypeScript**, **Node.js**, **FastAPI**, **PostgreSQL**, **REST APIs**, and **Docker** across hardware-software integrated engineering projects.
- Developed systems spanning **smart agriculture**, **smart waste management**, **smart city infrastructure**, **biometric IoT**, **IoT security**, and **blockchain-backed forensic evidence preservation**.
- Applied the complete IoT development lifecycle including **system architecture**, **hardware integration**, **sensor data acquisition**, **device communication**, **edge processing**, **API development**, **database integration**, **testing**, **debugging**, and **deployment**.

IoT Systems Engineering – Academic & Independent Projects

2024 – Present

- Designed and implemented **hardware-software integrated IoT prototypes** involving microcontrollers, sensors, actuators, communication protocols, monitoring systems, and automated control workflows.
- Developed practical solutions involving **soil health monitoring**, **automated irrigation**, **smart waste collection**, **smart-city infrastructure**, **RFID-based systems**, **biometric authentication**, and **connected-device telemetry**.
- Applied **MQTT**, **Wi-Fi**, **Bluetooth/BLE**, **REST APIs**, **embedded programming**, **real-time data processing**, **cloud connectivity**, and **database integration** to connect physical devices with software platforms.
- Integrated cybersecurity principles including **secure communication**, **authentication**, **cryptographic hashing**, **network security**, **threat modeling**, and **evidence integrity** into IoT and connected-system architectures.

Featured IoT Projects

Smart City IoT Street Light & Traffic Management System

ESP32, Embedded C/C++, MQTT,

Sensors, Edge Computing, IoT Security

- Designed a **secure Smart City IoT platform** integrating intelligent street lighting and dynamic traffic management using **ESP32**, **sensor-driven edge logic**, and **real-time telemetry**.
- Implemented **MQTT-based device communication with encrypted data transmission**, enabling reliable connectivity between distributed IoT devices and centralized monitoring services.
- Applied **edge computing**, **IoT security**, **threat modeling**, and **real-time monitoring** to support responsive and scalable smart-city infrastructure.

EcoBin – AI-Powered Smart Waste Management System

ESP32, IoT Sensors, Python, FastAPI, React,

TypeScript, XGBoost, OR-Tools

- Developed an **IoT-enabled smart waste management system** for real-time container monitoring, fill-level prediction, overflow detection, and intelligent collection planning.
- Integrated **ultrasonic**, **weight**, **gas**, and **RFID sensing technologies** with wireless connectivity to collect operational telemetry and support automated waste-management workflows.
- Built predictive services using **XGBoost** and **FastAPI** and applied **Google OR-Tools** for optimized collection routing.

Smart Irrigation System – IoT Soil Health & Automated Irrigation

Arduino Uno, ESP32, Embedded

C/C++, Soil Moisture, TMP36, Servo, DC Pump

- Engineered an **IoT-based smart irrigation prototype** for continuous soil and environmental monitoring with automated water-delivery control.
- Integrated **soil moisture**, **temperature**, and **light sensors** with Arduino, DC pump, servo actuator, LCD/I2C display, and optional ESP32 connectivity.
- Implemented **sensor data acquisition**, **threshold-based automation**, **actuator control**, and **real-time monitoring** to optimize irrigation and reduce unnecessary water consumption.

Smart Cart OS – AI-Powered Connected Shopping Cart

ESP32, RFID, IoT, TypeScript, React,

Computer Vision, Gemini AI

- Designed an **AI-enabled smart shopping cart system** combining IoT hardware concepts, RFID-based product identification, real-time analytics, and computer vision.

- Developed a connected-system architecture supporting **RFID simulation, real-time cart analytics, product identification, and intelligent shopping workflows.**
- Integrated **computer vision and Gemini AI** for automated product recognition, demonstrating the convergence of IoT, edge intelligence, and AI-powered systems.

ChainForensics – IoT Security & Forensic Evidence Platform ESP32, MQTT, Node.js, TypeScript, Ethereum, Solidity, IPFS, SHA-256

- Engineered an **IoT security and digital forensics platform** for collecting, verifying, and preserving security events generated by connected devices.
- Implemented an evidence pipeline using **IoT telemetry, MQTT event collection, deterministic serialization, SHA-256 hashing, blockchain smart contracts, and IPFS.**
- Designed mechanisms for **tamper detection, immutable evidence registration, chain-of-custody verification, and forensic timeline reconstruction** across distributed IoT environments.

Biometric Voting System – Secure IoT Authentication Platform ESP32, Mantra Fingerprint Device, TypeScript, React, Node.js, IoT Security

- Developing a **hardware-integrated biometric voting platform** using fingerprint authentication for secure and reliable voter identity verification.
- Upgraded the prototype from an **R307 fingerprint sensor to a Mantra fingerprint device**, strengthening hardware integration and real-world authentication workflows.
- Designed the system around **secure biometric verification, duplicate-vote prevention, auditability, device integration, and cybersecurity controls.**

Certifications

The Arduino Platform and C Programming – University of California, Irvine (Coursera)

Interfacing with the Arduino – University of California, Irvine (Coursera)

Introduction to Cyber Attacks – New York University (Coursera)

Data Structures Using Python: An Introduction – Coursera

Programming for Everybody (Getting Started with Python) – University of Michigan (Coursera)

Career Essentials in GitHub Professional Certificate – GitHub & LinkedIn Learning

Practical GitHub Actions – LinkedIn Learning

Practical GitHub Code Search – LinkedIn Learning

Practical GitHub Project Management and Collaboration – LinkedIn Learning

Object-Oriented Design – University of Alberta (Coursera)

Java Programming – Coursera

Quantum Computing for Everyone – Coursera

Nanotechnology and Nanosensors, Part 1 – Technion (Coursera)

Write Professional Emails in English – Georgia Institute of Technology (Coursera)

Relevant Coursework

IoT & Embedded Systems: Internet of Things, Embedded Systems, Microcontroller Programming, Sensor Networks, IoT Architecture, Hardware–Software Integration

Cybersecurity & Networking: Cybersecurity, Computer Networks, Network Security, Digital Forensics, Secure Communication, Information Security

Computer Science: Data Structures & Algorithms, Database Management Systems, Operating Systems, Object-Oriented Programming, Software Engineering

Engineering & Analytics: Probability & Statistics, System Design, Problem Solving, Technical Writing